

REQUEST FOR INFORMATION FOR LUNAR SCIENCE AND TECHNOLOGY PAYLOADS FOR EXPANDED LUNAR LANDING OPPORTUNITIES

Request for Information

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On March 24th, the National Aeronautics and Space Administration (NASA) will be unveiling an unprecedented approach to increase lunar landings to a near-monthly landing cadence starting in 2027 via the Commercial Lunar Payload Services (CLPS) initiative. This will dramatically expand the number of payloads flying to the Moon and unlock a drastic increase in science, technology and economic opportunities that will strongly benefit future exploration and humanity. This represents a transformative leap for commercial companies, universities, and research institutions in the U.S. and across the world. Humanity will have repeated and regular opportunities to get payloads manifested to land on the Moon across 10 CLPS deliveries per year.

This Request for Information (RFI) will inform how NASA approaches filling government-provided payload allocations on these numerous new CLPS deliveries in 2027, 2028 and beyond. If a payload is selected by NASA to fly on CLPS, NASA will pay for the launch, landing and surface mobility of payloads on the lunar surface and shall have its CLPS landing providers integrate payloads into their systems, with CLPS partners handling payload integration. CLPS will provide power, communications, thermal regulation, and potential survive-the-night capabilities, if a lander has a Radioisotope Heating Unit onboard. Payload providers will have multiple opportunities to manifest payloads aboard CLPS landers.

This RFI requests potential payload details that can be provided to NASA and potentially shared globally via an opt-in approach to a public Lunar Payload Database. Of greatest interest to NASA would be payload teams that are self-funded and can deliver their completed flight units to CLPS and perform their own operations while the payload is on the lunar surface. NASA's approach to working with payload providers will simplify the broader launch, landing and surface deployment approach for any payloads that aspire to land on the Moon.

Responding to this RFI via the NASA Solicitation and Proposal Integrated Review and Evaluation System (NSPIRES) system is open to people across the world. As NASA intends to share information about opted-in payloads via the Lunar Payload Database, this allows for broader interconnections in the growing lunar economy, connecting payload teams, funders, and CLPS delivery providers, among others.

NASA will evaluate RFI responses and additionally provide an option for RFI respondents to list their proposed payloads on a public database, including both payloads seeking flight opportunities as well as listing the payload sponsor.

This Lunar Payloads RFI seeks responses broadly from industry, universities, non-profit organizations, and individuals, including foreign entities; NASA centers, Federal Funded Research and Development Centers (FFRDCs) such as the Jet Propulsion Laboratory; and other U.S. federal, state, local, tribal government agencies. Responses are subject to the restrictions placed on NASA by SEC. 526 of the “Consolidated Appropriations Act, 2024” and all subsequent Acts that prohibit bilateral relationships with China. Details of that restriction are at: <https://www.congress.gov/118/plaws/publ42/PLAW-118publ42.pdf>

This National Aeronautics and Space Administration Request for Information (RFI) does not constitute a commitment, implied or otherwise, that the National Aeronautics and Space Administration (NASA) will take action in this matter.

Summary

On December 18, 2025, President Trump signed the Executive Order [*Ensuring American Space Superiority*](#), establishing a bold national directive to return Americans to the Moon by 2028 through the Artemis Program and to establish initial elements of a permanent lunar outpost by 2030. The Executive Order directs NASA to lead the world in space exploration, expand human presence in space, and lay the economic and scientific foundations for future missions to Mars — while simultaneously spurring at least \$50 billion in new commercial space investment by 2028.

National space policy prioritizes sending astronauts to the lunar surface and building a lunar outpost on the surface of the Moon: the Moon Base. To achieve this goal, the Commercial Lunar Payload Services (CLPS) approach will evolve to increase lander payload delivery capability, lander mission reliability and lander mission quantity and cadence. These changes will enable establishment of the Moon Base, large scale participation in Moon endeavors and unlock access to the lunar surface for many groups from industry, academia and government.

The CLPS initiative provides end-to-end delivery services to the lunar surface through U.S. commercial lander providers. CLPS serves as a critical and cost-effective pathway for delivering the science instruments, technology demonstrations, and infrastructure components needed to characterize the lunar environment, advance exploration capabilities, and enable a sustained human presence on the lunar surface.

This Lunar Payloads RFI comes as part of a broader release of NASA plans for an expanded approach to lunar exploration by 2028. This expanded approach includes a significant enhancement in the number of lunar landings per year using CLPS, with each landing providing space for payloads that advance Moon to Mars science and exploration goals. To enable planning of how to best utilize these additional CLPS flights, NASA is seeking information about payloads that could take advantage of additional CLPS flights.

Given the acceleration and increase in number of near-term CLPS flights, NASA has particular interest in the availability of *existing hardware* that can be ready for flight with little to no payload development necessary. Existing hardware is more likely to enable ‘ready-to-fly’ science and technology payloads that could take advantage of these new near-term opportunities for lunar landing and surface operations via CLPS. It is understood that some engineering efforts could be required to accommodate the payload for flight on specific CLPS-contractor landing craft.

NASA also seeks to take advantage of expanded CLPS landing opportunities after these near-term flights and is open to learning about payloads that require more development and would be ready in the medium term (~4-6 years). Suggested guidelines for payload accommodations are described in the Appendix called Summary of Prior CLPS Interface Guidance.

The Science Mission Directorate (SMD) is leading this RFI development and evaluation and will coordinate with the Exploration Systems Development Mission Directorate, Space Technology Mission Directorate and Space Operations Mission Directorate in its evaluation and distribution of RFI results. RFI results will inform NASA’s evolving plans for lunar activities, including CLPS and the SMD’s Lunar Discovery and Exploration Program. To re-emphasize, this RFI does not reflect a commitment that additional funding from these existing NASA programs for payloads will be forthcoming, nor that payload concepts submitted to the RFI will be manifested on CLPS deliveries.

Background and Risk Philosophy for Lunar Payloads

NASA's CLPS activities already target two lunar landings per year and has developed payloads to fill each of the currently planned landings. For past and current CLPS landings, NASA is in process of developing over 60 science and technology payloads to be delivered to the Moon to conduct research and technology demonstrations. These payloads cover a broad range of categories, including lander-mounted instruments and sensors, GPS receivers, position, navigation, and timing (PNT) tools, rovers, and other technologies that benefit from access to the surface of the Moon.

NASA has enabled a risk tolerant approach to payload development for flight on CLPS, recognizing the higher risk involved in commercial-led missions. NASA has managed its existing payloads to date under *NPR 7120.8 NASA Research and Technology Program and Project Management Requirements*. NASA does not mandate formal risk classification approaches for payloads per *NPR 8705.4B Risk Classification for NASA Payloads*.

Landing opportunities are generally expected to occur with stationary landers that are not expected to last beyond a single lunar day (14 Earth days). Landing sites will emphasize the South Pole region but will also include equatorial and mid-latitude landing sites, and far-side landings on the Moon. If a payload has a specific-requested landing region then it should be stated in the written input. Orbital deliveries are also potential places for deployment of CLPS payloads. Some existing CLPS task orders are planning to place assets into lunar orbit, and it is in-scope for the CLPS contract to deliver to that location, so, orbital payloads are also in-scope

for this RFI. Similarly, orbital payloads that could be incorporated on to Mars missions are also in-scope for this RFI.

Additional context on CLPS and interface requirements development

NASA's CLPS project coordinates and contracts with commercial vendors to land scientific and technology payloads on the lunar surface. CLPS is a delivery service and may carry NASA-sponsored and non-NASA payloads; thus, NASA expects to be one of many customers on a particular lunar surface delivery. CLPS deliveries are made of two components: instruments, such as those selected through NASA's Payloads for Research Investigations from the Surface of the Moon (PRISM), and Task Orders for lunar delivery generated through the CLPS office.

NASA will consider the results of this RFI as it develops a strategy for manifesting payloads onboard the expanded list of CLPS deliveries. NASA pays for the delivery of payloads that it selects and sponsors. If a payload is selected for delivery to the Moon by NASA, NASA will manifest it on a future CLPS landing. NASA's selected CLPS vendor will deliver the sponsored payload and other NASA-sponsored payloads to the lunar surface. All Task Orders are Firm Fixed Price for the full scope of delivery, which necessitates mature payload interfaces and requirements at the outset of the RFTP process. Several factors drive delivery costs, including, but not limited to, lander delivery schedule, landing site terrain, and accommodation (e.g., mass, power, communications, timing and duration of operations, special needs of the instruments, etc.) of the payloads. Therefore, payloads that grow to exceed proposed not-to-exceed (NTE) mass, resource use, or costs, or become at-risk to meeting the lander delivery schedule after selection will be at risk of being removed from the manifest for their CLPS delivery.

Requested Payload Description for Responses to RFI

RFI responses discussing payloads that can be manifested on CLPS are sought as part of this submission. Responses should discuss their potential lunar payload and answer the following questions via the NSPIRES cover page. Each question is limited to 1000 or 4000 characters in response (which can be roughly equivalent to 150 or 600 words), with length specified on the NSPIRES cover page. NASA is collecting the information using the cover page response sheet in order to facilitate easier evaluation of the responses. A separate and supplementary written document (PDF) of 5 pages in length is permitted to provide additional narrative, images or salient technical details that are not included in the below responses. NASA's primary assessment of RFI responses will be based on evaluating response to the below questions and it will only examine the supplementary PDF file if NASA seeks additional context beyond what is in the RFI response submissions. The PDF will not be shared beyond NASA.

Responses are requested along the following sections:

A. Entity Information:

- Q1. Responder's name
- Q2. Email contact information
- Q3. Institutional website
- Q4. Website for payload specific concept (if any)
- Q5. What is the payload's country of origin? Specify "USA only" or name the country or countries of origin
- Q6. Identify partners and Co-Investigators.

- Q7. Respondent Type (select all that apply):
 - i) Payload Developer
 - ii) Payload Funder
 - iii) Lunar Landing and Payload Delivery Service
 - iv) Surface Mobility Provider
 - v) Orbit Hosting Platform
- Q8. Respondent goal in responding to this RFI if other than proposing a payload
- Q9. Summary of previous relevant experience
- Q10. Opting in for inclusion of answers to Sections A, B, C, D, and E to public Lunar Payload database
 - Yes
 - No
- Q11. If opting out of inclusion in Lunar Payload Database, does recipient object to be publicly listed as a respondent to the RFI?
 - Yes
 - No

B. Payload Concept

- Q12. Payload name
- Q13. Payload concept description
- Q14. Planned timeline for delivery readiness (2027-2028) or medium term (2029-2032)
- Q15. Payload science objective(s) (if any)
- Q16. Payload technology objective(s) (if any)
- Q17. System-level ConOps description
- Q18. Primary Destination target (pick all that apply):
 - i) Equatorial regions of the Moon
 - ii) Mid-latitude regions
 - iii) Lunar South Pole region
 - iv) Landing site-agnostic (open to any region for landing)
 - v) Lunar near-side
 - vi) Lunar far-side
 - vii) Lunar orbit
 - viii) Mars orbit
 - ix) Other
- Q19. Additional details on desired deployment location (e.g., specific landing sites, or, if orbital, orbit characteristics and pointing requirements)

C. Technical Specifications of Your Payload:

- Q20. Mass (kg)
- Q21. Volume (cm x cm x cm)
- Q22. Communication requirements (volume, bandwidth, etc.)
- Q23. Communication interface (wired, wireless, interface port, etc.)
- Q24. Power Requirements 1: Nominal (W)
- Q25. Power Requirements 2: Peak (W)
- Q26. Power Requirements 3: Power conditioning needed or other special considerations
- Q27. Thermal conditioning requirements
- Q28. Sensitivity to radiation and electromagnetic interference/compliance
- Q29. Mechanical interface (thermal isolation, etc.)

- Q30. Optical sensitivities (dust, chemicals, line of sight, etc.)
- Q31. Expected payload acoustic and shock load limits

D. Moon to Mars Objectives:

- Q32. Describe Moon to Mars Objective, technology gap, data gap, and/or civil space shortfall that the payload supports. See the next section for background on these objectives, gaps and shortfalls. These serve as a description of NASA's desires for scientific and technology-focused work for lunar payloads.
- Q33. Select a primary objective
- Q34. List additional secondary objectives
- Q35. Provide a description of how the payload supports the primary and secondary objectives

E. Payload Heritage and Readiness

- Q36. Describe the heritage of your payload
- Q37. Describe the technology readiness level of your payload
- Q38. Describe what kind of effort would be required to make it flight-ready
- Q39. Describe any technical risks associated with delivering and integrating the hardware
- Q40. Describe any programmatic (cost or schedule) risks associated with delivering and integrating the hardware

F. Cost and Schedule:

- Q41. Description of remaining payload development costs
- Q42. Potential storage and sustainment costs
- Q43. Expected operations costs
- Q44. The timeline for completing development
- Q45. If opting into Lunar Payloads Database, do you separately want to include answers from this Section F in what is shared?
 - Yes
 - No

G. Business and Strategy Approach:

- Q46. RFI respondent's current plan or approach for funding the payload development, sustainment and operations
- Q47. Respondent's interest in providing multiple copies of the unit, including expected rebuild costs and timeline for follow-on units
- Q48. If opting into Lunar Payloads Database, do you separately want to include answers from this Section G in what is shared?
 - Yes
 - No

H. Additional Information for NASA

- Q49. Is there more information that you'd like to share with NASA?
- Q50. Are you a U.S. entity, as in an organization, or, if you are an individual respondent, do you meet the definition for U.S. person in Section §120.62 of Title 22 (Foreign Relations) of the Code of Federal Regulations. U.S. person (as defined in §120.62) means a person who is a lawful permanent resident as defined by 8 U.S.C. 1101(a)(20) or who is a protected individual as defined by 8 U.S.C. 1324b(a)(3)?

Note: If respondent opts into being publicly listed in the Lunar Payloads Database as part of Q10, responses to Sections F and G would not be made public to the NASA lunar payloads database unless separately noted in responses to Q45 and Q48.

Relevance to NASA: Background on NASA Moon to Mars objectives, technology gaps, data gaps, and civil space shortfalls

This section describes some of NASA's goals for Moon to Mars exploration so as to enable an RFI respondent to best describe how their payload ties to NASA's Moon to Mars Objectives and overall scientific and technical goals. Respondents to this RFI are required to identify the Moon to Mars objectives, technology gaps, data gaps, and/or technology shortfalls that their payload is relevant to. Respondents must identify a primary objective, gap, or shortfall that they support, and may note additional objectives, gaps or shortfalls that they also address. The response intake form for this RFI will require qualitative descriptions explaining why the proposed payload is relevant to the identified objectives.

NASA has made clear its interest in Moon to Mars exploration through the publication of the [Moon to Mars Objectives](#), which lay out a series of science, technology, infrastructure, and operational goals needed to expand humanity's presence into the solar system¹. The following are illustrative examples of how payloads may map to these objectives:

- Payloads that characterize and identify lunar volatiles — materials that can change state between gas, ice, and liquid depending on temperature — are relevant to LPS-03, which calls for revealing inner solar system volatile origin and delivery processes by determining the age, origin, distribution, abundance, composition, transport, and sequestration of lunar volatiles.
- Payloads that measure the solar wind from the lunar surface are relevant to HS-04, which seeks to improve understanding of magnetotail and pristine solar wind dynamics in the vicinity of the Moon.
- Payloads that monitor space weather and characterize the radiation and plasma environment at the lunar surface are relevant to HS-01, which seeks to improve understanding of space weather phenomena to enable enhanced observation and prediction of the dynamic environment from space to the surface at the Moon.
- Payloads that demonstrate or advance lunar surface power generation and distribution — such as small fission reactors, solar arrays, or energy storage systems — are relevant to LI-01, which calls for developing an incremental lunar power generation and distribution system evolvable to support continuous robotic and human operations.
- Payloads that test or advance lunar surface position, navigation, and timing (PNT) capabilities — such as lunar GPS receivers or navigation beacons — are relevant to LI-03, which calls for developing a lunar PNT architecture capable of scaling to support long-term science, exploration, and industrial needs.

¹ Information on the overall Moon to Mars Architecture can be found at this website: <https://www.nasa.gov/MoonToMarsArchitecture/>. A list of the NASA Moon to Mars objectives can be found here: [“NASA’s Moon to Mars Strategy and Objectives Document.”](#) An excel document that maps NASA’s M2M architecture to the M2M Objectives is [here](#).

- Payloads that demonstrate precision landing technologies are relevant to LI-05, which calls for demonstrating precision landing capabilities in support of continuous human lunar presence and a robust lunar economy.
- Payloads that demonstrate local or regional surface mobility — such as rovers or mobility platforms — are relevant to LI-06, which calls for demonstrating local, regional, and global surface transportation and mobility capabilities.
- Payloads that characterize, prospect for, or begin to demonstrate extraction of lunar resources — including water ice, regolith, or other in-situ materials — are relevant to AS-03, which calls for characterizing accessible lunar resources and enabling In-Situ Resource Utilization (ISRU), and to LI-07, which calls for demonstrating industrial-scale ISRU capabilities.
- Payloads that characterize the contemporary lunar surface environment — including micrometeorite flux, dust, and space weathering — are relevant to AS-01, which calls for characterizing and monitoring the contemporary environments of the lunar surface and orbit to plan and support crewed operations.

These examples are illustrative and not exhaustive. The Moon to Mars Objectives span planetary science, heliophysics, human and biological research, physical and fundamental science, applied science, lunar infrastructure, transportation and habitation, and operations. Respondents are encouraged to review the full Moon to Mars Objectives document and identify all objectives relevant to their proposed payload.

NASA's Technology Needs:

NASA's Space Technology Mission Directorate has released a set of [civil space shortfalls](#) that identify broad technology areas that require further development to meet future exploration, science, and other mission needs across the full scope of civil spaceflight. These shortfalls are further decomposed into a collection of need statements that highlight specific development objectives within the scope of each shortfall. Within the context of Moon to Mars exploration there is a direct one-to-one mapping between relevant need statements and the architecture-driven technology gaps outlined in NASA's Moon to Mars Architecture Definition Document (ADD, [Revision C updated in December 2025](#)). The ADD's Appendix D "Architecture-Driven Technology Gaps" provides additional detail and architectural context for these technologies that NASA has assessed as requiring further development to meet Moon to Mars objectives.

Prospective lunar technology payloads in response to this RFI are desired by NASA to be aligned to the agency's Moon to Mars objectives, civil space technology shortfalls, and M2M architecture-driven technology gaps.

Architecture-Driven Data Gaps: The Moon to Mars Architecture Definition Document ([Revision C updated in December 2025](#)) also identifies a series of data gaps where additional data is needed to enable required technology development or further exploration in support of Moon to Mars objectives. The ADD's Appendix E "Architecture-Driven Data Gaps" provides additional detail on these data needs within their Moon to Mars exploration context. Payloads that help provide data needed to close these gaps would also be of interest to NASA.

In elaboration of the Moon to Mars Objectives that cover science, there are additional resources to consider that describe science goals. Of special interest are the National Academies led decadal surveys that represent community consensus about the goals for a given domain of science. These surveys shaped the formation of the Moon to Mars objectives but consulting these source documents can provide additional context on science goals, opportunities and priorities. Such documents include, but are not limited to, the recently released [Origins, Worlds, and Life: A Decadal Strategy for Planetary Science and Astrobiology 2023-2032](#), [Thriving in Space: Ensuring the Future of Biological and Physical Sciences Research: A Decadal Survey for 2023-2032](#), [Pathways to Discovery in Astronomy and Astrophysics for the 2020s](#), and [A Decadal Strategy for Solar and Space Physics](#), as well as in the NAS [Scientific Context for Exploration of the Moon](#) and [A Science Strategy for the Human Exploration of Mars](#) (which includes precursor Mars science needs).

Public Database of Aspiring Lunar Payloads and Sponsoring Organizations

Across the world, a growing ecosystem of governments, research institutions, philanthropists, and commercial ventures is looking for ways to contribute to the next era of lunar exploration. As interest accelerates, early insight into potential future payload concepts becomes increasingly important for coordinating investments, aligning technical development, and ensuring that promising ideas do not evolve in isolation. International partners have already responded positively to NASA's Moon to Mars architecture and are actively pursuing systems and elements that can help advance shared exploration goals. Further, President Trump's recent executive order has signaled the possibility of significant new commercial funding for lunar activities, expanding the pool of organizations that may be able to support payload development and delivery. Sharing information about potential payloads enables multiple benefits, including connecting prospective funders to payload developers, linking payloads with launch providers and helping complementary scientific and technological payloads find one another to create more ambitious, integrated missions

NASA is establishing a publicly accessible Lunar Payloads Database to give the global lunar community a clearer view of the concepts, capabilities, and interests emerging worldwide. Respondents to this RFI may opt in to having their payload concepts listed in the database, enabling CLPS companies, international partners, philanthropists, and other stakeholders to easily see what potential missions are being envisioned and how they align with NASA's broader Moon to Mars objectives

Opting into the public database is voluntary; NASA will consider all RFI responses, whether listed publicly or not, as it shapes strategies and plans for expanding CLPS flight opportunities to the lunar surface.

Organizations or individuals interested in funding future lunar payloads, either by supporting their development or paying for their delivery through CLPS, may also register in the database (see questions Q7 and Q8 in the [Requested Payload Description for Responses to RFI](#)). This helps payload developers identify prospective sponsors and increases the likelihood that promising concepts can secure the resources needed to reach the Moon.

Current CLPS providers, as well as emerging lunar lander companies, may request to be listed in the database. They will have access to the public payload entries, allowing them to understand what types of payloads are in development, how those payloads may influence lander design choices, and where private arrangements for lunar delivery may be possible. Scientific mission proposers can also use the database to identify complementary investigations and discover technologies that may strengthen future proposals.

In addition, any group or individual seeking to fund, support, or fly lunar payloads is welcome to list their information so they can be easily identified by prospective payload teams and collaborators.

NASA does not endorse any of the posting entities, nor do we warrant the accuracy of the information or viability of the proposed payloads. NASA reserves the right not to list respondents in the Lunar Payload Database per its discretion.

Specific Technology Areas of Interest for Lunar Science

NASA's technology interests are captured in the above Moon to Mars objectives, technology gaps, and civil space shortfalls. For developing lunar science needs, NASA is also interested in technologies that enable:

- Survive and Operate through the Lunar Night
- RPS and nuclear-enabled power and survive the night technologies.
- Freezer Technology for Lunar Cold-Conditioned Sample Return,
- Rapid and Autonomous Long-Distance Lunar Rover Traversing

Post RFI Response Submission Approach

Following this initial feedback, as part of its market research, NASA may conduct one-on-one meetings with respondents. These meetings will allow for exchange of information and will provide an opportunity for respondents to provide feedback on the Government's preliminary requirements and acquisition approach. Once any subsequent solicitations are released, no further conversations will take place until that solicitation is completed.

Multiple RFI responses from an individual entity are acceptable if they are exploring multiple payload concepts. Most RFI responses are expected to be framed around a specific payload concept and thus submissions from a single entity focused on multiple payloads would be of value.

NASA may conduct meetings with respondents such as topical workshops and invite relevant stakeholders such as venture capitalists and philanthropists to attend the meeting. These completely voluntary meetings provide an opportunity for respondents to provide more detailed feedback to NASA and for private sector participants working on payloads to engage with each other and to potentially create new options for payloads that might fly on CLPS. Inclusion of relevant stakeholders may also help create new collaborations and opportunity for payloads to land and operate on the Moon.

Any public discussion of the results of this RFI may disclose information about the identities of the respondents unless the RFI respondents request to remain anonymous via Question 11.

It is not NASA's intent to disclose publicly Respondents' proprietary information obtained in response to this RFI. To the full extent that it is protected pursuant to the Freedom of Information Act and other laws and regulations, information identified by a Respondent as "Proprietary or Confidential" will be kept confidential. When deciding whether to mark RFI responses questions as being submittable to the above Section Public Database of Aspiring Lunar Payloads and Sponsoring Organizations, respondents should not include proprietary responses as being slated for public consumption.

It is emphasized that this RFI is NOT a Request for Proposal, nor is it an Invitation for Bid. This RFI is being used to obtain information for planning purposes only, and the Government does not intend to award a contract at this time. As stipulated in FAR 15.201(e), responses to this notice are not considered offers and cannot be accepted by the Government to form a binding contract. Pursuant to FAR 52.215-3, entitled Request for Information or Solicitation for Planning Purposes, this information is being made available for market research, information, and planning purposes and to allow the broader community the opportunity to verify reasonableness and feasibility of the requirement, as well as promote competition. This RFI is subject to review or cancellation at any time and is not to be construed as a commitment by the Government to enter into a contract. The Government will not pay for the information submitted in response to this request.

Please do not request a copy of the solicitation, as no solicitation exists at this time. If a firm requirement is developed and a solicitation is issued, the solicitation will be made available through NSPIRES (<https://nspires.nasaprs.com/external/>). It is the responsibility of Offerors and interested parties to monitor the internet sites for the release of the solicitation and amendments, if any, and they will be responsible for downloading their own copy of the documents. NASA Clause 1852.215-84, Ombudsman, is applicable. The Center Ombudsman for potential acquisitions can be found at <https://www.hq.nasa.gov/office/procurement/regs/Procurement-Ombuds-Comp-Advocate-Listing.pdf> The Ombudsman for this RFI will be the Deputy Assistant Administrator for Procurement.

Requested Information

Section Requested Payload Description for Responses to RFI described the information that NASA is requesting from respondents to the RFI. This section clarifies aspects of how the respondent and NASA should prepare and handle the information.

On a strictly voluntary basis, respondents may submit confidential information, controlled unclassified information (CUI), proprietary information, or export-controlled information, including International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR) restricted information, appropriately and clearly identified as such, in response to this RFI. If such is submitted, the respondent should ensure that their response is not marked as being for inclusion in the public database. A respondent can still

submit an RFI response for inclusion in the public database but then choose to provide CUI/ITAR/EAR information in the separate PDF attachment. The PDF attachment should be appropriately marked.

Note that it is the respondent's responsibility to determine whether any submitted information is subject to the provisions of ITAR or EAR. Information about U.S. export regulations is available at https://www.pmddtc.state.gov/ddtc_public/ddtc_public and at <https://www.bis.gov/>.

While inclusion of export-controlled material in the submission is not prohibited, respondents are advised that NASA may employ the services of non-U.S. persons (individuals who are neither U.S. citizens nor lawful permanent residents of the U.S. or U.S. citizens employed by a foreign organization) to process responses submitted in response to this RFI. In order to enable proper processing, any export-controlled information subject to ITAR or EAR must be marked with a notice to that effect. If the response contains export-controlled material, the material shall be presented in a red font or enclosed in a red-bordered box, and the following statement shall be prominently displayed at the top of the first page of the submitted file:

“The information (data) contained in [insert page numbers or other identification] of this response is (are) subject to U.S. export laws and regulations. It is furnished to the Government with the understanding that it will not be exported without the prior approval of the respondent under the terms of an applicable export license or technical assistance agreement. The identified information (data) is (are) printed in a red font and figure(s) and table(s) containing the identified information (data) is (are) placed in a red-bordered box.”

NASA also requests and collects a limited set of demographic data from all registered NSPIRES users for the purpose of future analysis of the demographic differences associated with NASA's award and review processes. Demographic items requested include sex, race, ethnicity, and disability status. Submission of these demographic data are voluntary, confidential, and not a precondition for using NSPIRES or submitting a response to this RFI. If you do not wish to provide demographic responses, choose the buttons labeled “I Prefer Not to Answer”.

Response Instructions

All responses must be received by 11:59 p.m. Eastern time on April 23rd, 2026.

All responses to this RFI must be submitted in an electronic format via the NASA Solicitation and Proposal Integrated Review and Evaluation System (NSPIRES), located at <http://nspires.nasaprs.com/>.

For this RFI, responses involve responding to the questions included on the NSPIRES cover page, noted in Section Requested Payload Description for Responses to RFI above. For this RFI, the core response submission will take the form of a Notice of Intent (NOI) within the NSPIRES online announcement data management system. The RFI response itself will be a series of Program Specific Data to be entered into the NOI Cover Page, addressing the questions shown

above. No PDF-formatted file needs to be attached to the NOI. Respondents can provide a supplementary 5 page pdf response that provides important context that the respondent wishes NASA to be aware of. However, NASA's primary assessment of the RFI will be based on the responses to the Program Specific Data. NASA will only examine the supplementary PDF file if NASA seeks additional context beyond what is in the RFI response submissions. The PDF will not be shared beyond NASA.

All PDF responses must adhere to the following formatting requirements:

Length:	5 pages maximum
Page size:	8.5" x 11.0" paper size
Paragraph:	Single-spaced, single-column text with no more than 5.5 lines per vertical inch of text
Margins:	One-inch margins on all four sides with no content in the margins
Font:	Font size 12, not to exceed 15 characters per horizontal inch, including spaces, sans serif font recommended.
Inserts:	Figures, tables, and other inserts are permitted

You must be registered with NSPIRES to submit a RFI response. See registration instructions at <http://nspires.nasaprs.com/> (select "Create an account"). Neither institution registration nor an institution affiliation is required to respond to this RFI.

When you have an NSPIRES account, go to the NSPIRES page for Request For Information for Lunar Science and Technology Payloads for Expanded Lunar Landing Opportunities

1. Click "Create"
2. Log in when requested
3. Enter the Request for Information: Response title NOTE: NSPIRES will show this as "NOI title"
4. Select "do not link at this time" for submitting organization page.
5. Click "Save" on next page.

Note: It is required to complete these "Program Specific Data" questions on the NSPIRES cover page called the "NOI Details". You must have text in each required box. Do not leave any required text box blank. You may use N/A if you have nothing to say about a required answer. Special characters such as formulae are not supported by the text box.

7. Click Submit NOI button. NOTE: Clicking "Submit NOI" does not complete the submission process. Clicking "Submit" will take you to the NOI submission confirmation page, which provides you with the NOI/RFI number for your records.

Please note: You may delete and replace form fields and uploaded documents any time before the submission deadline, however once your RFI is submitted, it cannot be deleted. Unsubmitted responses will not be distributed.

6. Point of Contact

Please email questions and comments concerning this RFI to the email address below with the subject line: Lunar Payloads RFI. Depending on the nature of received questions, NASA may respond on an individual basis by email or may post responses to inquiries in a “Questions and Answers” document available on NSPIRES. Any posted Q&As will be edited to preserve the anonymity of persons and institutions who submit questions and are intended to address inquiries of broader interest and general clarification.

HQ-RFILunarPayloads@mail.nasa.gov

The below emails are not to be used for RFI response submissions; any submissions made via these email addresses will not be considered.

The following officials are overseeing this RFI and its evaluation. They can be reached using the email address above.

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Appendix: Summary of Prior CLPS Interface Guidance to Inform Payload Developers

CLPS Payload Accommodation Guidance

The following is payload accommodation guidance drawn from recent CLPS task order experience, including a recent [Payload and Research Investigations from the Surface of the Moon \(PRISM\) call](#). These will likely apply to many of the expanded CLPS landing opportunities but the exact technical approaches for those landings are still in formulation. As such, the following is described as guidance – but not requirements -- that can help RFI respondents understand the complexities of potentially manifesting a payload on CLPS. Addressing these interface accommodations in detail is not being requested as part of this RFI response but may inform responses to the above section Requested Payload Description for Responses to RFI, specifically to questions about technical and programmatic risk. If an RFI respondent sees challenges in some of these areas they can note it as a risk there.

There is no mass limit for potential payload responses to this RFI. It is expected that some new CLPS landings will have payload capacities on the orders of >>>100kg. Payloads that are smaller than that mass are also acceptable and more easily accommodable by CLPS landers. It is possible that some near-term CLPS deliveries may be able to deliver significantly higher masses to the surface.

Payload concepts that will be explored for delivery via CLPS must be accommodated by landing service vendors who will be contracted through NASA's Commercial Lunar Payload Services (CLPS) master contract. The commercial service vendor will be competitively selected and, in most cases, a potential payload must be lander/CLPS vendor agnostic and utilize common interfaces where possible.

General Tenets for Proposed Payloads

The following describes design philosophy that CLPS payloads have generally followed and that respondents should be aware of as context for the complexity of manifesting on CLPS. If a payload is manifested on CLPS, the payload will need to demonstrate alignment with these tenets:

- Use common and contemporary standards (see below for examples).
- Reduce operational complexity (e.g., number of operational modes, number of mode transitions, etc.).
- Employ a 20% or more mass margin
- Reduce demand on resources, especially at critical times.
 - For operations during launch and descent mission phases, avoid any resource requests beyond survival unless the payload's priority purpose is to operate during those mission phases.
- Avoid the need for any non-essential operation/monitoring/access.
- Respondents should be aware that downlink data rates are shared across the lander and all payloads, are vendor- and task order-specific, and can be a significant accommodation driver. If manifested on CLPS, a payload will need to clearly describe total data volume needed and the timeline for when data would need to be on the ground — for example: a constant stream of data, a picture that immediately needs to be downloaded, or a multi-hour measurement that can be downloaded after completion.
- Document clear contamination requirements for each instrument, as needed.
- Respondents should be aware that CLPS payloads have typically been designed to meet MIL-STD-461G, RE102 and CE102, with a 3-second dwell and RE testing below 100MHz not required. If a payload is manifested on CLPS, compliance with applicable EMC standards will be required.
- Respondents should be aware that CLPS payloads have typically been designed to meet GSFC-STD-7000 General Environments Verification Standard for:
 - Random Vibe, protoflight or Sine vibe (if fundamental frequency is <100Hz)
 - Thermal vacuum (4 cycles)

If a payload is manifested on CLPS, compliance with applicable environmental verification standards will be required.

- Respondents should be aware that SLSSPEC-159 Rev I, Cross Program Design Specification for Natural Environments (DSNE) has been used as source data for expected lunar environments in recent CLPS task orders.
- If needed resources are mode-dependent, payloads will need to be very clear about usage peaks, phasing, averages, durations, etc.
- If a payload is manifested on CLPS, the payload will need to include handling requirements for proposed instruments and any protective covers or special mechanical

ground support equipment, as necessary, along with a description of the impact to science objectives if these requirements are not met.

- If a payload is manifested on CLPS, the payload will need to minimize risk to other payloads and the CLPS lander, and document hazards associated with doing harm to other payloads or the CLPS lander.
- Launch licensing and launch site requirements: The CLPS vendor will be compliant to all safety requirements imposed by the Range, the U.S. Federal Aviation Administration, and others. If a payload is manifested on CLPS, the payload will need to support identification and mitigation of hazards and provide supporting documentation to the CLPS provider, including:
 - Verifications to Space Force Space Command Manual (AFSPCMAN) 91-710 V3 dated 2022.
 - Battery compliance documentation in accordance with applicable safety standards. Respondents should be aware that battery hazards are a significant area of scrutiny in the CLPS manifest process.
- If a payload is manifested on CLPS, the payload will need to avoid hazardous materials, designs, and operations where possible, and shall provide hazard analyses, mitigations, and documentation support as needed.
- Respondents should be aware that CLPS payloads have typically used Grade C or D EEE parts.

Payload Design Details

The list below represents the fuller set of interface and concept of operations (conops) documentation that would be required if a payload is manifested on CLPS. It is provided here as a preview of future rigor and to help respondents understand the complexity of the CLPS manifest process — it is not an RFI ask.

If a payload is manifested on CLPS, the payload will need to document expected mass and dimensions, and as many interface and conops requirements as possible, with special consideration given to the following, if applicable:

- Payload thermal conditioning requirements (operating/non-operating temperature limits, survival heater power, lander/payload thermal interface, etc.)
- Payload communication interfaces and formats (wired, wireless, etc.)
- Payload power requirements (number of power channels, nominal, peak, power conditioning, etc.) and any grounding requirements
- Payload pointing, field-of-view, or orientation requirements
- Payload configuration on the lander requirements (e.g., deployment on the end of a boom, deployment from lander, distance from the lunar surface, etc.), inclusive of pyrolytic and non-pyrolytic activation devices, and how they would be activated
- Payload mechanical interface(s) and launch lock requirements, if any
- Payload optical sensitivities (dust, chemicals, line-of-sight to the Sun, etc.) and keep out zones for sensor operation, if applicable
- Payload cleanliness requirements (e.g., organic contamination, particulates, magnetic cleanliness, etc.)
- Payload-unique time synchronization and time stamp requirements
- Payload operational timeline (operation during transit, operation during lunar orbit, etc.)

- Special handling and storage requirements (i.e., GN2 purge — note how long experiment(s) can be off purge and if it's needed at the lander facility, pad, and while encapsulated)
- Payload conops considerations (e.g., cruise phase/descent phase operations, local time operational constraints)

Thermal:

- Mounting – Adiabatic (Acceptable – must specify radiation FOV)
- Mounting – Conductive (Acceptable – must specify dissipation)
- If a payload is manifested on CLPS, the payload will need to state the allowable temperature range and how much energy will be transferred across the instrument/lander interface as a function of the interface temperature. Payloads that push more energy across that interface are generally less accommodable than those that minimize thermal transfer across the interface.